

TOPIC 1: INTRODUCTION TO GENETICS & MENDEL'S PEA EXPERIMENTS

PU-II Principles of Inheritance and Variation • Topic-Wise Worksheet

Student Name: _____

Roll / Reg. No: _____

Date / Batch: _____

SECTION A: Multiple Choice Questions (1 Mark Each)

- Which of the following is a dominant trait in garden pea (*Pisum sativum*)?
(A) Violet flower colour (B) Terminal flower position
(C) Constricted pod shape (D) Yellow pod colour
- Mendel conducted his hybridization experiments on garden peas for how many years?
(A) 5 years (1850–1855) (B) 7 years (1856–1863)
(C) 10 years (1860–1870) (D) 4 years (1865–1869)
- The degree by which progeny differs from its parents is defined as:
(A) Inheritance (B) Heredity
(C) Variation (D) Polymorphism
- Sahiwal cows developed in Punjab through artificial selection originate from ancestral:
(A) Wild sheep (B) Wild ancestral cows
(C) Domestic buffalo (D) Jersey cows

SECTION B: Fill in the Blanks (1 Mark Each)

- The physical site or specific location of a gene on a homologous chromosome is known as
- Genes coding for a pair of contrasting traits are slightly different forms of the same gene called
- Mendel selected true-breeding pea plant lines as 7 contrasting pairs for his breeding studies.
- The external physical appearance of an organism is termed its

SECTION C: Very Short Answer Questions (2 Marks Each)

9. [2M] Define 'Homozygous' and 'Heterozygous' conditions with suitable symbolic examples.

10. [2M] Give two major reasons why Gregor Mendel selected garden pea (**Pisum sativum**) for his hybridization experiments.

SECTION D: Short Answer Questions (3 Marks Each)

11. [3M] List any three contrasting pairs of traits studied by Mendel in pea plants, mentioning both dominant and recessive forms.

12. [3M] Define the terms: (a) Back cross (b) Test cross (c) Out cross.

SECTION E: Long Answer / Essay Questions (5 Marks Each)

13. [5M] State five key reasons for Mendel's experimental success and discuss why pea plants served as an ideal material for genetics research.

14. [5M] Explain the basic terminology of genetics: Gene, Allele, Phenotype, Genotype, and Dominant vs. Recessive trait with suitable illustrations.